

Cloud-Based Energy Consumption Analytics using AWS S3, Snowflake & Tableau

1. Introduction

This project demonstrates an end-to-end cloud-based data analytics workflow using AWS S3, Snowflake, SQL, and Tableau. The main objective of the project is to analyze household energy consumption patterns across different countries, regions, and energy sources while generating business insights through interactive dashboards.

The project covers the complete analytics pipeline starting from data storage in AWS S3, cloud data warehousing in Snowflake, SQL-based data transformation, and dashboard visualization in Tableau.

2. Objectives

- The primary objectives of this project are:
- To create a cloud-based analytics pipeline.
 - To integrate AWS S3 with Snowflake using Storage Integration.
 - To load and transform energy consumption data using Snowflake SQL.
 - To analyze household energy usage and cost savings.
 - To create interactive dashboards in Tableau for business insights.

3. Technologies Used

Technology Purpose

AWS S3	Cloud storage for dataset
Snowflake	Cloud data warehouse
SQL	Data transformation and analysis
Tableau	Data visualization and dashboarding

4. Dataset Description

The dataset contains household energy consumption information from multiple countries and regions.

Dataset Columns

Column Name	Description
Household_ID	Unique household identifier
Region	Geographic region

Column Name	Description
Country	Country name
Energy_Source	Type of energy used
Monthly_Usage_kWh	Monthly electricity consumption
Year	Data year
Household_Size	Number of family members
Income_Level	Income category
Urban_Rural	Area classification
Adoption_Year	Renewable adoption year
Subsidy_Received	Government subsidy status
Cost_Savings_USD	Savings generated

5. Project Architecture

CSV Dataset



AWS S3 Bucket



Snowflake External Stage



Snowflake SQL Transformations



Tableau Dashboard

6. AWS S3 Data Storage

The dataset CSV file was uploaded into an AWS S3 bucket. The bucket acts as a cloud storage layer that allows Snowflake to access the data securely using Storage Integration.

Steps Performed

1. Created AWS S3 bucket.
 2. Uploaded CSV dataset into the bucket.
 3. Configured IAM Role for Snowflake access.
 4. Established secure integration between Snowflake and AWS S3.
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7. Snowflake Integration

A Storage Integration object was created in Snowflake to securely connect AWS S3 with Snowflake.

Storage Integration SQL

```
CREATE OR REPLACE STORAGE INTEGRATION tableau_Integration
TYPE = EXTERNAL_STAGE
STORAGE_PROVIDER = 'S3'
ENABLED = TRUE
STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::195969197650:role/tableau.role'
STORAGE_ALLOWED_LOCATIONS = ('s3://tableau4.project/')
COMMENT = 'Optional Comment';
```

8. Database and Table Creation

A database, schema, and table were created in Snowflake for storing the dataset.

SQL Queries

```
CREATE DATABASE tableau;

CREATE SCHEMA tableau_Data;

CREATE TABLE tableau_dataset (
Household_ID STRING,
Region STRING,
Country STRING,
Energy_Source STRING,
Monthly_Usage_kWh FLOAT,
Year INT,
Household_Size INT,
Income_Level STRING,
Urban_Rural STRING,
Adoption_Year INT,
Subsidy_Received STRING,
Cost_Savings_USD FLOAT
);
```

9. Creating External Stage

An external stage was created to access the dataset stored in AWS S3.

SQL Query

```
CREATE STAGE tableau.tableau_Data.tableau_stage
URL = 's3://tableau4.project'
STORAGE_INTEGRATION = tableau_Integration;
```

10. Loading Data into Snowflake

The COPY INTO command was used to load CSV data from AWS S3 into Snowflake.

SQL Query

```
COPY INTO tableau_dataset
FROM @tableau_stage
FILE_FORMAT = (TYPE = CSV FIELD_DELIMITER = ',' SKIP_HEADER = 1)
ON_ERROR = 'CONTINUE';
```

11. Data Transformation using Snowflake SQL

Several SQL transformations were performed on the dataset to simulate business analysis scenarios.

Monthly Usage Updates

```
UPDATE energy_consumption
SET monthly_usage_kwh = monthly_usage_kwh * 1.1
WHERE income_level = 'Low';
```

```
UPDATE energy_consumption
SET monthly_usage_kwh = monthly_usage_kwh * 1.2
WHERE income_level = 'Middle';
```

```
UPDATE energy_consumption
SET monthly_usage_kwh = monthly_usage_kwh * 1.3
WHERE income_level = 'High';
```

Cost Savings Updates

```
UPDATE energy_consumption
SET cost_savings_usd = cost_savings_usd * 0.9
WHERE income_level = 'Low';
```

```
UPDATE energy_consumption
SET cost_savings_usd = cost_savings_usd * 0.8
WHERE income_level = 'Middle';
```

```
UPDATE energy_consumption
SET cost_savings_usd = cost_savings_usd * 0.7
WHERE income_level = 'High';
```

12. Analytical SQL Queries

Region-wise Energy Consumption

```
SELECT
    region,
    SUM(monthly_usage_kwh) AS total_usage
FROM energy_consumption
```

```
GROUP BY region
ORDER BY total_usage DESC;
```

Energy Source Analysis

```
SELECT
    energy_source,
    AVG(cost_savings_usd) AS avg_savings
FROM energy_consumption
GROUP BY energy_source;
```

13. Tableau Dashboard

The transformed Snowflake data was connected to Tableau for visualization and dashboard creation.

Dashboard Features

- KWH by Country
- KWH by Region
- KWH by Energy Source
- Cost Savings by Region
- Cost Savings by Energy Source

The dashboard provides interactive insights into:

- Energy consumption trends
 - Regional comparisons
 - Energy source performance
 - Cost-saving analysis
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14. Key Insights

- Wind energy showed the highest energy consumption.
 - Europe recorded high overall energy usage.
 - Higher-income households consumed more electricity.
 - Renewable energy adoption influenced cost savings positively.
 - Regional variations significantly impacted energy consumption patterns.
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15. Challenges Faced

- Configuring AWS IAM roles correctly.
- Establishing Snowflake Storage Integration.

- Managing CSV data loading errors.
 - Designing an organized Tableau dashboard layout.
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16. Conclusion

This project successfully demonstrates a complete cloud-based analytics workflow integrating AWS S3, Snowflake, SQL, and Tableau. The project highlights practical experience in cloud storage, data warehousing, SQL transformations, and business intelligence dashboarding.

The solution provides meaningful business insights into household energy consumption and renewable energy adoption trends while showcasing modern data analytics skills relevant for Data Analyst and Business Intelligence roles.

17. Future Improvements

- Add real-time streaming data integration.
- Implement advanced SQL analytics.
- Add Tableau filters and KPI cards.
- Integrate predictive analytics using Python.
- Deploy dashboards publicly using Tableau Cloud.

18. Author

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